

Literature Benchmark Brief: Wang_2025_ApplEnergy

Title: Uncertainty quantification and dynamic operating reserve sizing in renewable-dominated grids

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Cluster: Cluster D: Probabilistic Sizing & Reserve

Core Theoretical Mechanism & Model Formulation

Dynamic conditional reserve procurement based on state-dependent renewable uncertainty

Empirical Scale & Experimental Dataset

Regional utility power system with high wind-solar penetration

Relevance to IEEE TSTE & Sustainable Energy Systems

Shows that static reserve margins waste capital; reserves must vary with wind state

Identified Gaps & Operational Vulnerabilities

Assumed perfect communication infrastructure without SCADA packet dropouts

Role in Manuscript Defense & Reviewer #2 Proof

Supports state-conditional quantile reserve formulation in Section II-D